Quantification: 535_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	20
interior lacuna count	15
frame-edge lacuna count	5

Interior lacunae only: mean, median and sample SD (n = 15)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2953.47	2501	1734.82	px^2
major_axis_length_px	lacuna shape	84.6058	84.6082	27.5467	px
minor_axis_length_px	lacuna shape	47.1388	41.4719	17.579	px
aspect_ratio	lacuna shape	1.8754	1.7243	0.505	unitless
eccentricity	lacuna shape	0.7975	0.8147	0.1316	unitless
solidity	lacuna shape	0.8169	0.8161	0.0905	unitless
roots_count	headline	8.2667	8	3.1728	count
ring_length_r30_px	headline	281.4	259	84.0483	px
ring_length_r60_px	reported	599.133	616	192.637	px
owned_length_px	ownership-dependent	864.323	680.671	575.641	px
edge_count	ownership-dependent	38	24	24.8251	count
mean_edge_length_px	ownership-dependent	23.5211	21.8226	7.385	px
perimeter_px	normalised	284.739	262.149	111.782	px
ring_area_r30_px2	normalised	9229.2	9278	2400.49	px^2
ring_area_r60_px2	normalised	21088.1	21092	4777.18	px^2
in_frame_fraction_r30	normalised	0.9889	1	0.0428	unitless
in_frame_fraction_r60	normalised	0.982	1	0.0686	unitless
ring_density_r30	normalised	0.0308728	0.0316015	0.00637931	px^-1
ring_density_r60	normalised	0.0283163	0.0292054	0.00596646	px^-1
roots_per_100px_perimeter	normalised	3.0359	3.0579	0.9355	per 100 px
ring_attached_length_r30_px	network v2	247.6	226	85.4992	px
ring_attached_length_r60_px	network v2	448.133	465	180.92	px
ring_length_w_r30_px	network v2	319.49	298.593	95.7512	px
ring_length_w_r60_px	network v2	680.779	689.257	219.261	px
sholl_crossings_r10	network v2	6.9333	6	2.5486	count
sholl_crossings_r20	network v2	8.8	9	3.7455	count
sholl_crossings_r30	network v2	7.9333	8	3.2834	count
loop_count	loops, report only	1.7333	0	2.4631	count
loop_edge_share	loops, report only	0.1066	0	0.1285	share
ring_width_mean_r30_px	width	7.7619	7.7833	0.9764	px

Per field
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measure	value	units
analysed_area_px2	1.00017e+06	px^2
total_skeleton_length_px	22854	px
canalicular_length_density_per_px	0.0228501	px^-1
junction_count	378	count
junction_density_per_px2	0.00037794	px^-2
lacuna_count	20	count

lacunae_per_px2	2e-05	px^-2
skeleton_component_count	385	count
mean_component_length_px	59.361	px
median_component_length_px	21	px
field_length_density_w_per_px	0.025735	px^-1
field_density_without_flagged_per_px	0.0222666	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	586	px
unattached_component_count	36	count
canalicular_length_density_attached_per_px	0.0228501	px^-1
field_length_density_with_unattached_per_px	0.023436	px^-1
loop_count	29	count
loop_edge_share	0.0974	share
edge_count	1129	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.6248	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	19710	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	766	55.3886	19.9792	2.7723	0.9327	0.8815	1.5239	7.094	208.97	True
2	1417	59.2424	36.6108	1.6182	0.7862	0.7182	0.2448	34.1404	249.047	False
3	492	40.0308	17.439	2.2955	0.9001	0.8571	-1.4757	7.689	940.191	True
4	3380	83.9727	56.9872	1.4735	0.7345	0.8272	-0.9852	92.8941	716.001	False
5	532	31.4193	21.8844	1.4357	0.7175	0.9449	-0.5795	222.887	852.859	False
6	1982	85.686	32.1789	2.6628	0.9268	0.8251	-1.1473	243.598	120.229	False
7	670	34.0595	26.8161	1.2701	0.6165	0.9293	0.3429	300.875	1012.39	True
8	1956	84.6082	30.6094	2.7641	0.9323	0.8815	-0.4607	416.865	210.624	False
9	7294	147.859	76.1316	1.9421	0.8573	0.7458	-0.0651	474.999	284.411	False
10	4856	115.922	57.0373	2.0324	0.8706	0.8161	-1.3108	461.194	574.536	False
11	3706	78.2414	69.9705	1.1182	0.4475	0.789	0.5114	581.897	502.58	False
12	2088	68.5717	40.0641	1.7115	0.8116	0.9198	0.595	570.565	144.556	False
13	2662	104.173	41.4719	2.5119	0.9173	0.6642	1.1534	613.184	873.72	False
14	2280	82.4791	35.9781	2.2925	0.8998	0.9205	-1.1889	632.433	317.15	False
15	1049	49.7107	28.8293	1.7243	0.8147	0.9468	1.1742	770.511	315.037	False
16	4401	92.4669	72.9809	1.267	0.6141	0.7373	1.4761	819.541	677.56	False
17	4198	98.8193	64.1556	1.5403	0.7606	0.7155	1.4566	847.201	603.76	False
18	2501	85.9153	42.1925	2.0363	0.8711	0.8016	1.0022	846.978	318.042	False
19	1473	60.4844	33.1719	1.8234	0.8362	0.8731	-0.2053	858.547	1008.1	True
20	703	38.461	24.103	1.5957	0.7793	0.9083	-0.1158	967.855	1012.53	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	0	51	210	0	0	0	
2	6	209	354	393.433	23	17.1058	
3	4	81	264	183.882	7	26.2689	8.1954
4	7	351	834	680.671	23	29.5944	7.4729
5	5	203	544	525.973	23	22.8684	7.5355
6	9	398	928	2019	96	21.0313	8.2992
7	1	61	362	200.024	10	20.0024	10.932
8	7	259	488	676.5	31	21.8226	9.2021
9	16	427	813	1139.9	62	18.3855	8.3815
10	11	290	651	872.382	34	25.6583	8.3643
11	11	391	733	1015.9	46	22.0849	7.96
12	7	227	616	354.865	19	18.6771	6.293
13	10	298	692	2233.84	78	28.6389	9.4006
14	7	229	366	330.735	17	19.455	8.07
15	2	152	334	746.441	16	46.6525	7.6066
16	9	229	450	492.049	24	20.502	6.4983
17	9	209	421	426.037	19	22.423	6.2141
18	8	349	763	1057.12	59	17.9173	7.3678
19	5	140	272	725.529	41	17.6958	7.4904
20	4	99	229	43.9706	4	10.9926	7.8076

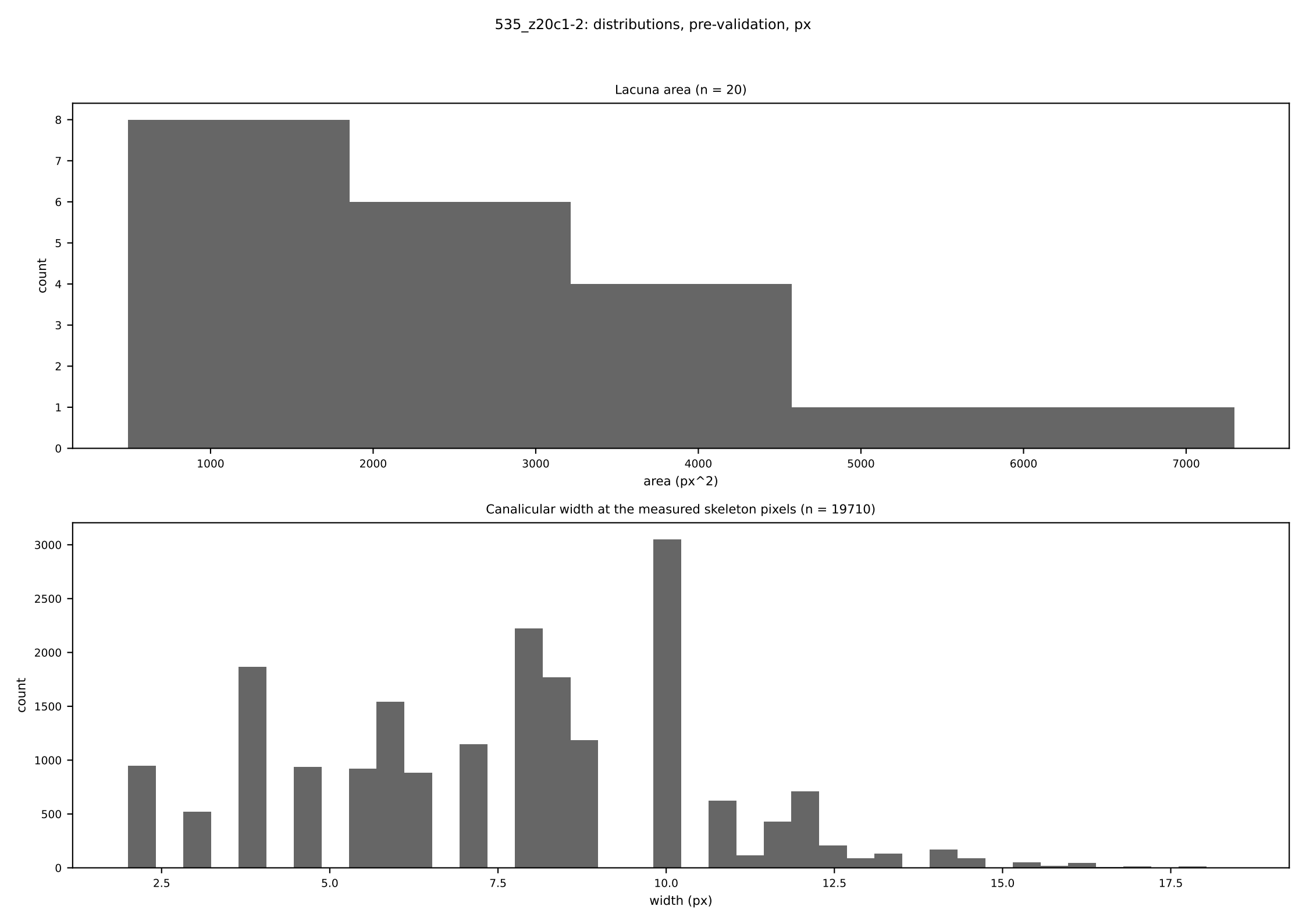
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	153.326	2172	4611	0.5445	0.5303	0.0234807	0.0455433	0
2	226.072	5807	13167	0.8341	0.734	0.035991	0.0268854	2.654
3	101.941	3528	9806	0.6148	0.5733	0.0229592	0.0269223	3.9238
4	303.605	10283	25801	1	0.9954	0.034134	0.0323243	2.3056
5	92.8112	5363	16377	1	1	0.037852	0.0332173	5.3873
6	243.179	9488	24489	1	1	0.0419477	0.0378946	3.701
7	106.184	3521	9851	0.5977	0.5656	0.0173246	0.0367475	0.9418
8	208.894	7431	16405	1	1	0.034854	0.029747	3.351
9	446.174	13512	24714	1	1	0.0316015	0.0328963	3.586
10	391.931	12275	26488	1	1	0.0236253	0.0245772	2.8066
11	312.191	11238	24088	1	1	0.0347927	0.0304301	3.5235
12	198.995	8411	21092	1	1	0.0269885	0.0292054	3.5177
13	327.019	11681	28795	1	1	0.0255115	0.0240319	3.0579
14	223.965	8975	20144	1	1	0.0255153	0.0181692	3.1255
15	139.681	5912	13025	1	1	0.0257104	0.025643	1.4318
16	473.073	9278	19872	1	1	0.024682	0.0226449	1.9025
17	421.339	9660	21532	1	1	0.0216356	0.0195523	2.136
18	262.149	9124	20333	1	1	0.0382508	0.0375252	3.0517
19	173.539	4894	10893	0.636	0.5937	0.0286065	0.0249702	2.8812
20	112.912	3718	7367	0.6229	0.5173	0.0266272	0.0310846	3.5426

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	11	11	57.7696	240.38	1	0	4
2	189	253	238.995	404.534	9	5	4
3	75	105	90.0122	293.037	3	1	5
4	330	651	400.676	954.992	8	14	8
5	184	476	239.279	627.985	4	9	8
6	375	815	452.433	1047.05	11	10	11
7	61	135	66.6274	413.505	1	2	1
8	259	449	298.593	563.63	6	7	9

9	353	563	478.99	918.394	8	16	14
10	250	480	320.593	731.612	12	9	7
11	332	663	445.99	846.666	8	13	12
12	125	217	255.693	689.257	6	7	10
13	220	465	343.948	781.583	6	11	10
14	226	356	256.551	415.505	6	8	6
15	90	198	172.782	372.421	2	7	5
16	223	293	258.823	518.345	7	3	3
17	209	293	226.711	467.363	6	3	3
18	349	550	402.291	872.352	5	10	9
19	139	252	164.196	309.451	3	3	6
20	74	78	108.012	252.409	2	1	2


535_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 20)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 19710)
count
width (px)